Exercise Sheet 26

The problems use the conventions of Lecture 26: ε-coordinates, Embeddings I and II for the orthogonal chain, and the ordered symplectic basis (e_1, e_2, f_1, f_2) of $\mathbb{C}^4$. The interlacing rules for $B_n \to D_n$ and $D_n \to B_{n-1}$ and the symplectic branching theorem are the cited theorems of Goodman–Wallach, Section 8.1.1. Exercises 26.1–26.2 practise dimensions under restriction. Exercises 26.3–26.4 treat embedding dependence and multiplicities greater than one. Exercises 26.5–26.6 reconstruct and transfer the proofs.

Exercise 26.1 (The adjoint chain). Let $\lambda = (1, 1, 0)$, the highest weight of the adjoint module $\Lambda^2\mathbb{C}^7$ of $\mathfrak{so}_7$.

(i) Restrict $L(\lambda)$ to $\mathfrak{so}_6$ and then to $\mathfrak{so}_5$ using the interlacing rules. Give all dimensions.

(ii) Check the final answer by decomposing $\Lambda^2(\mathbb{C}^5 \oplus \mathbb{C}^2)$, where $\mathfrak{so}_5$ acts trivially on $\mathbb{C}^2$.

Exercise 26.2 (A half-integral restriction). Let $\lambda = \omega_1 + \omega_3 = (\frac{3}{2}, \frac{1}{2}, \frac{1}{2})$ for B_3.

(i) Prove $\dim L(\lambda) = 48$ by the Weyl dimension formula.

(ii) Restrict $L(\lambda)$ to $\mathfrak{so}_6$ and verify the dimensions.

(iii) Restrict the D_3-summand $L(\frac{3}{2}, \frac{1}{2}, \frac{1}{2})$ to $\mathfrak{so}_5$ and verify its dimension again.

Exercise 26.3 (Embedding dependence in $\mathfrak{sp}_4$). Let h_s and h_ℓ be the coroots of the root copies of $\mathfrak{sl}_2$ for $\varepsilon_1 - \varepsilon_2$ and $2\varepsilon_1$. Their matrices in the basis (e_1, e_2, f_1, f_2) are

$$D(h_s) = \operatorname{diag}(1, -1, -1, 1).$$

$$D(h_\ell) = \operatorname{diag}(1, 0, -1, 0).$$

(i) Show that h_s and h_ℓ act on a weight vector of weight $a\varepsilon_1 + b\varepsilon_2$ by $a - b$ and a.

(ii) Restrict $L(\omega_2) = \Lambda^2_0\mathbb{C}^4$ to both copies.

(iii) Restrict the adjoint module $L(2\omega_1)$ to both copies.

(iv) Explain the short-root answers through $\mathfrak{so}_3 \subset \mathfrak{so}_5$.

Exercise 26.4 (Multiplicity two from $\mathfrak{sp}_6$ to $\mathfrak{sp}_4$). Use the symplectic branching theorem.

(i) Restrict $L(1, 0, 0) = \mathbb{C}^6$ to $\mathfrak{sp}_4$.

(ii) Restrict $L(1, 1, 0) = \Lambda^2_0\mathbb{C}^6$, of dimension 14, to $\mathfrak{sp}_4$.

(iii) Check (ii) directly by writing $\mathbb{C}^6 = \mathbb{C}^4 \oplus P$, with P the fixed plane.

Exercise 26.5 (The parity operator and the rank-one model). Use the exterior model of Lecture 26, with $\theta = \gamma(e_0)$. Its Clifford relations, quadratic action and Cartan computation also hold for $n = 1$: their proofs use only the pairwise anticommutation relations.

(i) Prove that θ commutes with $\rho(X)$ for every $X \in \mathfrak{so}(W)$, and that it anticommutes with $\rho(\varphi(e_0 \wedge w))$ for every $w \in W$.

(ii) Deduce again that $\Lambda^{\mathrm{ev}}E$ and $\Lambda^{\mathrm{odd}}E$ are $\mathfrak{so}(W)$-submodules, and that $\rho(\varphi(e_0 \wedge w))$ maps each into the other.

(iii) For $n = 1$, let $A = \varphi(e_0 \wedge e_1)$, $B = \varphi(e_0 \wedge f_1)$ and $H = \varphi(e_1 \wedge f_1)$. Prove $[A, B] = -H$ in $\mathfrak{so}_3$. Compute the matrices $D(A)$, $D(B)$ and $D(H)$ of $\rho(A)$, $\rho(B)$ and $\rho(H)$ in the basis $(1, e_1)$ of $\Lambda\mathbb{C}e_1$, and verify the same relation.

Exercise 26.6 (Two steps at once). Let $n \geq 2$ and let λ be a B_n-dominant integral weight. Restrict $L(\lambda)$ from $\mathfrak{so}_{2n+1}$ to $\mathfrak{so}_{2n-1}$ through $\mathfrak{so}_{2n}$, using Embeddings I and II.

(i) Let ν be B_{n-1}-dominant and have the same type as λ. Prove from the interlacing rules that the multiplicity of $L(\nu)$ is the number of D_n-dominant μ of the same type as λ interlacing between λ and ν.

(ii) Show that the spin module restricts to $2S'$, using the propositions of Lecture 26 only.

(iii) For $k \in \mathbb{Z}_{\geq 0}$, show that the trivial module occurs $k+1$ times in the restriction of $L(k, 0, \ldots, 0)$. Conclude that the two-step restriction is not multiplicity-free for $k \geq 1$.